



PYTHON BOOTCAMP

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SETS



Sets:

- A set of data within curly bracket {...}
- Unordered collection of elements

SETS



Sets operations:

```
3  fruits = {"apple", "banana", "orange"}
4  numbers = {1, 2, 3, 4, 5}
5
6  # Set operations
7  fruits.add("grape")           # Add element
8  fruits.remove("banana")      # Remove element
9  fruits.discard("kiwi")       # Remove if exists (no error)
10
11 print(fruits)
```

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Sets mathematic operations:

```
14 set1 = {1, 2, 3, 4}
15 set2 = {3, 4, 5, 6}
16
17 print(set1.union(set2))      # {1, 2, 3, 4, 5, 6}
18 print(set1.intersection(set2)) # {3, 4}
19 print(set1.difference(set2))  # {1, 2}
```

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Exercise:

1. Create a system that stores student grades as tuples (name, subject, grade) and uses sets to find unique subjects and students.

```
grades = [  
    ("Alice", "Math", 85),  
    ("Bob", "Science", 92),  
    ("Alice", "Science", 78),  
    ("Charlie", "Math", 90),  
    ("Bob", "Math", 88),  
    ("Alice", "English", 95)  
]
```